

SERVICIO NACIONAL DE APRENDIZAJE – SENA
Centro de Tecnología de la Manufactura Avanzada (CTMA)

Actividad Proyecto FullStack Docker

Aprendiz:
Manuela Córdoba Robledo

Ficha:
3229209

Programa de Formación:
Análisis y Desarrollo de Software (ADSO)

Instructor:
Richard Alexander Betancur Sierra

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1. Descripción del proyecto

Este proyecto es una aplicación full-stack para gestionar tareas, compuesta por tres partes que corren en contenedores Docker: un backend en Express y MySQL que expone una API REST, una base de datos MySQL con una tabla de tareas, y un frontend en Astro con React que muestra la interfaz al usuario. Docker Compose orquesta los tres servicios, levantándolos en el orden correcto (primero la base de datos, luego el backend, luego el frontend), y un servidor Nginx en el frontend actúa como proxy hacia la API del backend.

2. Historial de commits

Commit 1: chore: inicializar repositorio del proyecto

```
Pc@LAPTOP-1K0GKBQD MINGW64 ~/Downloads/fullstack-docker/fullstack-docker
$ git init
git add README.md
git commit -m "chore: inicializar repositorio del proyecto"
Initialized empty Git repository in C:/Users/Pc/Downloads/fullstack-docker/fullstack-docker/.git/
warning: in the working copy of 'README.md', LF will be replaced by CRLF the next time Git touches it
[master (root-commit) d2e4938] chore: inicializar repositorio del proyecto
1 file changed, 28 insertions(+)
create mode 100644 README.md

Pc@LAPTOP-1K0GKBQD MINGW64 ~/Downloads/fullstack-docker/fullstack-docker (master)
$ AC

Pc@LAPTOP-1K0GKBQD MINGW64 ~/Downloads/fullstack-docker/fullstack-docker (master)
$ git add backend
git commit -m "feat(backend): crear API REST con Express y MySQL"
warning: in the working copy of 'backend/.env.example', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'backend/Dockerfile', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'backend/package.json', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'backend/src/index.js', LF will be replaced by CRLF the next time Git touches it
[master 805697e] feat(backend): crear API REST con Express y MySQL
```

Commit 2: feat(backend): crear API REST con Express y MySQL

```
Pc@LAPTOP-1K0GKBQD MINGW64 ~/Downloads/fullstack-docker/fullstack-docker (master)
$ git add backend
git commit -m "feat(backend): crear API REST con Express y MySQL"
warning: in the working copy of 'backend/.env.example', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'backend/Dockerfile', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'backend/package.json', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'backend/src/index.js', LF will be replaced by CRLF the next time Git touches it
[master 805697e] feat(backend): crear API REST con Express y MySQL
4 files changed, 114 insertions(+)
create mode 100644 backend/.env.example
create mode 100644 backend/Dockerfile
create mode 100644 backend/package.json
create mode 100644 backend/src/index.js
```

Commit 3: feat(db): agregar script de inicializacion con tabla y datos

```
Pc@LAPTOP-1K0GKBQD MINGW64 ~/Downloads/fullstack-docker/fullstack-docker (master)
$ git add db
git commit -m "feat(db): agregar script de inicializacion con tabla y datos"
warning: in the working copy of 'db/init.sql', LF will be replaced by CRLF the next time Git touches it
[master 6f49bea] feat(db): agregar script de inicializacion con tabla y datos
1 file changed, 14 insertions(+)
create mode 100644 db/init.sql
```

Commit 4: feat(frontend): creacion de interfaz con Astro y componentes React para las tareas

```
Pc@LAPTOP-1K0GKBQD MINGW64 ~/Downloads/fullstack-docker/fullstack-docker (master)
$ git add frontend
git commit -m "feat(frontend): creacion de interfaz con Astro y componentes React para las tareas"
warning: in the working copy of 'frontend/Dockerfile', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'frontend/astro.config.mjs', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'frontend/nginx.conf', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'frontend/package.json', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'frontend/src/components/TaskList.jsx', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'frontend/src/pages/index.astro', LF will be replaced by CRLF the next time Git touches it
[master 6750d24] feat(frontend): creacion de interfaz con Astro y componentes React para las tareas
6 files changed, 125 insertions(+)
create mode 100644 frontend/Dockerfile
create mode 100644 frontend/astro.config.mjs
create mode 100644 frontend/nginx.conf
create mode 100644 frontend/package.json
create mode 100644 frontend/src/components/TaskList.jsx
create mode 100644 frontend/src/pages/index.astro
```

Commit 5: feat(docker): creacion de Dockerfiles multistage para frontend y backend

```
Pc@LAPTOP-1K0GKBQD MINGW64 ~/Downloads/fullstack-docker/fullstack-docker (master)
$ git add backend/Dockerfile frontend/Dockerfile frontend/nginx.conf
git commit -m "feat(docker): creacion de Dockerfiles multistage para frontend y backend"
On branch master
Untracked files:
  (use "git add <file>..." to include in what will be committed)
    COMMITS.md
    docker-compose.yml

nothing added to commit but untracked files present (use "git add" to track)
```

Commit 6: feat(docker): creacion de configuracion Docker Compose para MySQL, backend y frontend

```
Pc@LAPTOP-1K0GKBQD MINGW64 ~/Downloads/fullstack-docker/fullstack-docker (master)
$ git add docker-compose.yml
git commit -m "feat(docker): creacion de configuracion Docker Compose para MySQL, backend y frontend"
warning: in the working copy of 'docker-compose.yml', LF will be replaced by CRLF the next time Git touches it
[master c490399] feat(docker): creacion de configuracion Docker Compose para MySQL, backend y frontend
1 file changed, 47 insertions(+)
create mode 100644 docker-compose.yml

Pc@LAPTOP-1K0GKBQD MINGW64 ~/Downloads/fullstack-docker/fullstack-docker (master)
$ git add docker-compose.yml
git commit -m "feat(docker): creacion de configuracion Docker Compose para MySQL, backend y frontend"
On branch master
Untracked files:
  (use "git add <file>..." to include in what will be committed)
    COMMITTS.md

nothing added to commit but untracked files present (use "git add" to track)
```

Commit 7: docs: agregar guía de commits del proyecto

```
Pc@LAPTOP-1K0GKBQD MINGW64 ~/Downloads/fullstack-docker/fullstack-docker (master)
$ git add COMMITTS.md
git commit -m "docs: agregar guía de commits del proyecto"
warning: in the working copy of 'COMMITTS.md', LF will be replaced by CRLF the next time Git touches it
[master 2f78d06] docs: agregar guía de commits del proyecto
1 file changed, 50 insertions(+)
create mode 100644 COMMITTS.md
```

3. Evidencia de ejecución

Captura del comando docker compose up --build finalizando correctamente, y captura de la aplicación funcionando en el navegador.

```
Pc@LAPTOP-1K0GKBQD MINGW64 ~/Downloads/fullstack-docker/fullstack-docker (master)
$ docker compose up --build
failed to connect to the docker API at npipe:////./pipe/dockerDesktopLinuxEngine; check if the path is correct and if the daemon is running: open //./pipe/dockerDesktopLinuxEngine: The system cannot find the file specified.

Pc@LAPTOP-1K0GKBQD MINGW64 ~/Downloads/fullstack-docker/fullstack-docker (master)
$ AC

Pc@LAPTOP-1K0GKBQD MINGW64 ~/Downloads/fullstack-docker/fullstack-docker (master)
$ docker compose up --build
[+] up 14/14
  ✓ Image mysql:8.0 Pulled                                17.4s
[+] Building 29.7s (29/29) FINISHED
=> [internal] load local bake definitions                    0.0s
=> => reading from stdin 1.15kB                             0.0s
=> [backend internal] load build definition from Dockerfile 0.0s
=> => transferring dockerfile: 356B                          0.0s
=> [frontend internal] load build definition from Dockerfile 0.0s
=> => transferring dockerfile: 418B                           0.0s
=> [frontend internal] load metadata for docker.io/library/node:22-alpine 1.4s
=> [frontend internal] load metadata for docker.io/library/nginx:alpine 1.5s
```

localhost:8080

App Full Stack con Docker

Mis tareas

Nueva tarea

- Configurar CD
- Hacer commits
- Realizar proyecto

4. Repositorio en GitHub

La URL de tu repo:

<https://github.com/robledocordoba123-cmyk/fullstack-docker>

```
Pc@LAPTOP-1K0GKBQD MINGW64 ~
$ cd ~/Downloads/fullstack-docker/fullstack-docker

Pc@LAPTOP-1K0GKBQD MINGW64 ~/Downloads/fullstack-docker/fullstack-docker (master)
$ git remote add origin https://github.com/robledocordoba123-cmyk/fullstack-docker.git
git branch -M main
git push -u origin main
info: please complete authentication in your browser...
Enumerating objects: 33, done.
Counting objects: 100% (33/33), done.
Delta compression using up to 12 threads
Compressing objects: 100% (28/28), done.
Writing objects: 100% (33/33), 7.27 KiB | 1.82 MiB/s, done.
Total 33 (delta 4), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (4/4), done.
To https://github.com/robledocordoba123-cmyk/fullstack-docker.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.

Pc@LAPTOP-1K0GKBQD MINGW64 ~/Downloads/fullstack-docker/fullstack-docker (main)
$ |
```

Commit 8-feat: agregar endpoints PUT/DELETE y aplicar tema morado al frontend

Se implementaron los dos endpoints adicionales solicitados por el instructor:

- PUT /api/tasks/:id → actualiza título y estado de una tarea
- DELETE /api/tasks/:id → elimina una tarea

```
Pc@LAPTOP-1K0GKBQD MINGW64 ~/Downloads/fullstack-docker/fullstack-docker (main)
$ git push
Enumerating objects: 17, done.
Counting objects: 100% (17/17), done.
Delta compression using up to 12 threads
Compressing objects: 100% (7/7), done.
Writing objects: 100% (9/9), 2.26 KiB | 2.26 MiB/s, done.
Total 9 (delta 4), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (4/4), completed with 4 local objects.
To https://github.com/robledocordoba123-cmyk/fullstack-docker.git
 2f78d06..aacc89 main -> main
```

También se rediseñó el frontend con un tema morado/azul, agregando botones para marcar tareas como completadas y eliminarlas.

5. Endpoints adicionales

Por instrucción del profesor, se implementaron dos endpoints adicionales al CRUD básico: **PUT /api/tasks/:id** para actualizar una tarea (marcarla como completada) y **DELETE /api/tasks/:id** para eliminarla. Ambos se probaron desde la interfaz web, agregando botones de check y borrado.

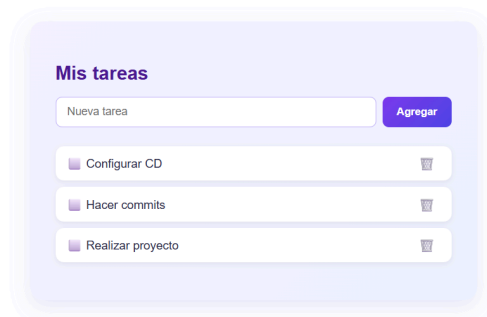
```
Restricted Mode is intended for safe code browsing. Trust this folder to enable all features. Manage Learn More

Explorer
  fullstack-docker
  fullstack-docker
  backend
  src
    JS index.js
  .env.example
  Dockerfile
  package.json
  db
  init.sql
  frontend
  src
    components
    TaskList.jsx
    pages
  astro.config.mjs
  Dockerfile
  nginx.conf
  package.json
  COMMITS.md
  docker-compose.yml
  README.md

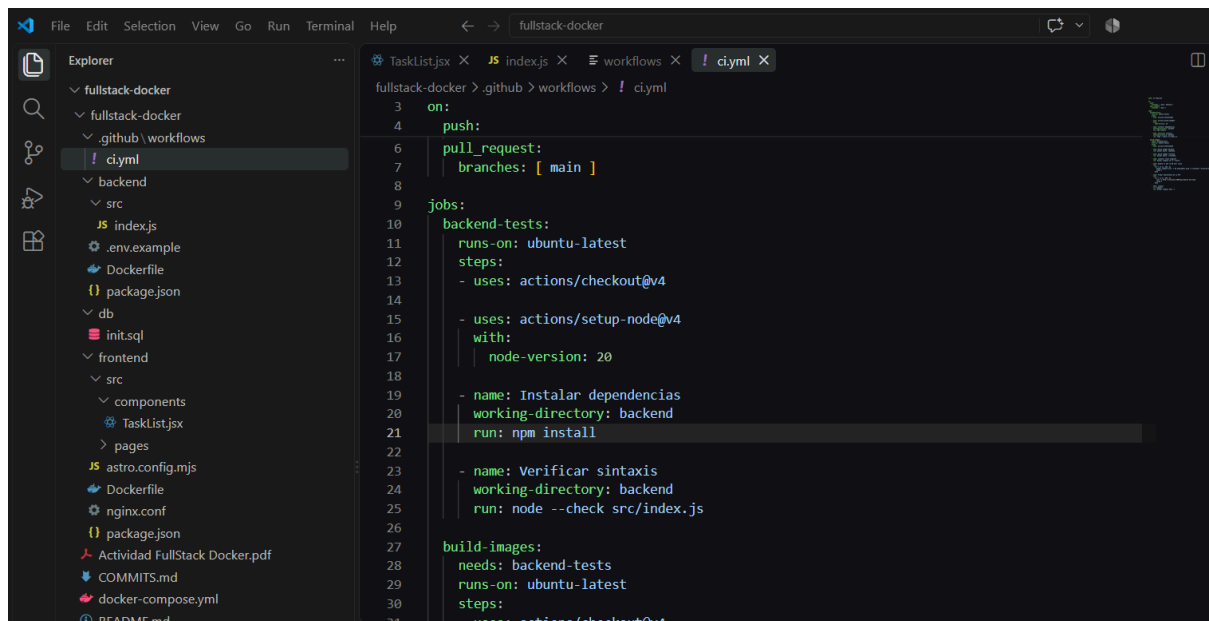
fullstack-docker > backend > src > JS index.js
43 app.post('/api/tasks', async (req, res) => {
44   // ...
45   // ...
46   // ...
47   // ...
48   // ...
49   // ...
50   // ...
51   // ...
52   // ...
53   // ...
54   // ...
55   // ...
56   // --- Actualizar una tarea (título y/o estado completado) ---
57   app.put('/api/tasks/:id', async (req, res) => {
58     try {
59       const { id } = req.params;
60       const { title, completed } = req.body;
61       await pool.query(
62         'UPDATE task SET title = ?, completed = ? WHERE id = ?',
63         [title, completed, id]
64       );
65       res.json({ id, title, completed });
66     } catch (error) {
67       res.status(500).json({ status: 'error', message: error.message });
68     }
69   });
70   // --- Eliminar una tarea ---
71   app.delete('/api/tasks/:id', async (req, res) => {
72     try {
73       const { id } = req.params;
74       await pool.query('DELETE FROM task WHERE id = ?', [id]);
75       res.json({ message: 'Tarea eliminada', id });
76     } catch (error) {
77       res.status(500).json({ status: 'error', message: error.message });
78     }
79   });
80   // ...
81   // ...
82   const port = process.env.PORT || 3000;
83   app.listen(port, () => {
84     console.log(`API en http://localhost:${port}`);
85   });
86 }
```



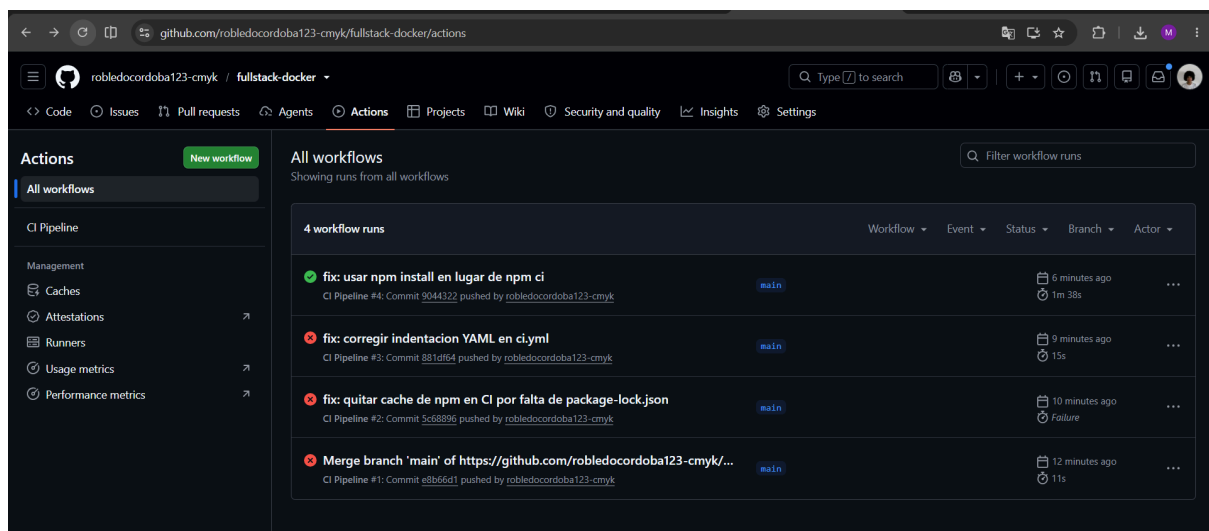
App Full Stack con Docker



6. Pipeline CI/CD



```
3  on:
4    push:
5
6  pull_request:
7    branches: [ main ]
8
9  jobs:
10    backend-tests:
11      runs-on: ubuntu-latest
12      steps:
13        - uses: actions/checkout@v4
14
15        - uses: actions/setup-node@v4
16          with:
17            node-version: 20
18
19        - name: Instalar dependencias
20          working-directory: backend
21          run: npm install
22
23        - name: Verificar sintaxis
24          working-directory: backend
25          run: node --check src/index.js
26
27    build-images:
28      needs: backend-tests
29      runs-on: ubuntu-latest
30      steps:
31        - uses: actions/checkout@v4
```



github.com/robledocordoba123-cmyk/fullstack-docker/actions

robledocordoba123-cmyk / fullstack-docker

Code Issues Pull requests Agents Actions Projects Wiki Security and quality Insights Settings

Actions New workflow

All workflows Filter workflow runs

Showing runs from all workflows

4 workflow runs

	Workflow	Event	Status	Branch	Actor
✅ fix: usar npm install en lugar de npm ci	CI Pipeline #4: Commit 9044322 pushed by robledocordoba123-cmyk	main	6 minutes ago 1m 38s		...
❌ fix: corregir indentacion YAML en ci.yml	CI Pipeline #3: Commit 881d64 pushed by robledocordoba123-cmyk	main	9 minutes ago 15s		...
❌ fix: quitar cache de npm en CI por falta de package-lock.json	CI Pipeline #2: Commit 5c58896 pushed by robledocordoba123-cmyk	main	10 minutes ago Failure		...
❌ Merge branch 'main' of https://github.com/robledocordoba123-cmyk/...	CI Pipeline #1: Commit e8b66d1 pushed by robledocordoba123-cmyk	main	12 minutes ago 11s		...

Por instrucción del instructor, se integró el archivo de configuración **ci.yml** proporcionado por él, ubicado en **.github/workflows/**. Este pipeline ejecuta automáticamente pruebas del backend y construcción de las imágenes Docker cada vez que se hace push a la rama principal.

